

Hamed Asadi

MSc Environmental Engineering | Prospective PhD Researcher

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Research Profile

I am an environmental researcher specializing in spatial modelling and data-driven conservation systems. My work has developed from GIS-based spatial decision-support and suitability analysis toward climate-biodiversity modelling using R-based analytical workflows. I integrate species distribution modelling (SDMs), CMIP6 climate scenarios, and structured decision-support frameworks to generate decision-relevant insights for biodiversity conservation and environmental management.

Research Interests

- Spatial environmental modelling and applied geospatial analysis
- Species distribution modelling (SDMs) and climate change impacts
- Climate scenario assessment (CMIP6; SSP pathways)
- Conservation planning and protected area assessment
- Structured spatial decision-support (MCDA) for environmental management

Education

- **MSc**, Natural Resources Engineering – Environmental Sciences — Isfahan University of Technology, Iran (2013–2016)

Thesis: Evaluating the ecotourism attractions of Abbas Abad Wildlife Refuge

Methods: GIS–MCDA (DANP, OWA), spatial suitability mapping

GPA: 15.88/20 | **Thesis Grade:** 19.62/20

- **BSc**, Natural Resources Engineering – Environmental Sciences — University of Guilan, Iran (2009–2013)

Project: Analysis of five production wastes in Songhor city

GPA: 15.83/20 | **Project Grade:** 19/20

Publications

Ahmadi, M., Nawaz, M. A., Asadi, H., et al. (2024). Protecting alpine biodiversity in the Middle East from climate change: Implications for high-elevation birds. Diversity and Distributions.

Asadi, H.*, Soffianian, A., Hemami, M.-R., et al. (2022). A hybrid GIS-OWA and DANP method for the identification and evaluation of ecotourism attractions. *GeoJournal* (Springer). (*Corresponding author; **MSc thesis-derived study**)

Research Experience

- Climate-driven biodiversity impact assessment using SDM workflows and CMIP6 scenarios in R-based environments.
- Development of GIS-based suitability and prioritization maps integrating spatial and environmental datasets.
- Identification of climate refugia and GIS-based protected area gap analysis for conservation planning.
- Design and implementation of structured MCDA decision frameworks (DANP–OWA).
- Translation of modelling outputs into decision-relevant spatial planning and management scenarios.
- Application of R for data analysis and modelling in collaborative ecological research projects.

Technical & Computational Skills

- R / RStudio: data manipulation, visualization, and environmental modelling workflows; experience in SDM-related analyses.
- Python (basic data processing).
- ArcGIS: spatial modelling, suitability mapping, environmental layer processing.
- Climate scenarios: CMIP6 (SSP2-4.5, SSP5-8.5).
- MCDA tools: DEMATEL, DANP, OWA.

Professional Experience

Environmental Consultant — Bina Azma Sepahan Company (Laboratory Set Consulting Engineers / Knowledge-Based) (Nov 2013 – Present)

Employment timeline: Full-time (44 hrs/week) Nov 2013 – Feb 2017 · Part-time (25 hrs/week) Feb 2017 – Dec 2017 · Full-time (44 hrs/week) Dec 2017 – Present

Key responsibilities

- EIA & environmental consulting for industrial projects and urban development plans.
- Environmental compliance & procedures: reviewing and improving protection instructions and workflows.
- Risk & pollution assessment: identifying pollution sources using water/soil sampling and lab results.

- Waste & wastewater management: recommendations for industrial waste disposal and treatment solutions.
- Regulatory consulting: guidance on environmental regulations and organizational requirements.
- Reporting & decision support: interpreting test outputs and translating them into actionable management solutions.

Academic Service

Peer Reviewer for international peer-reviewed journals in environmental science and ecology.

ORCID: 0000-0003-0435-5733

English Proficiency

PTE Academic: Overall, 80 (Speaking 90, Writing 74) — Valid until Dec 2027

Honors & Awards

Ranked 18th among >3300 participants in the Iranian National MSc Entrance Exam (2013).

Selected as an Exceptionally Talented Student in the undergraduate program (2013).

Certifications & Training

- Data Manipulation in R (Intersect Australia, 2022).
- Data Visualization in R (Intersect Australia, 2022).
- R Programming Software (Matris Pedia Training, 2022).
- Learn to Program Python (Intersect Australia, 2022).
- GIS Software (University of Guilan, 2013).
- ENVI Software (Isfahan University of Technology, 2013).
- Health, Safety & Environment (HSE) course (Isfahan University of Technology, 2015–2016).

Languages

Kurdish (native), Persian (native), English (fluent).